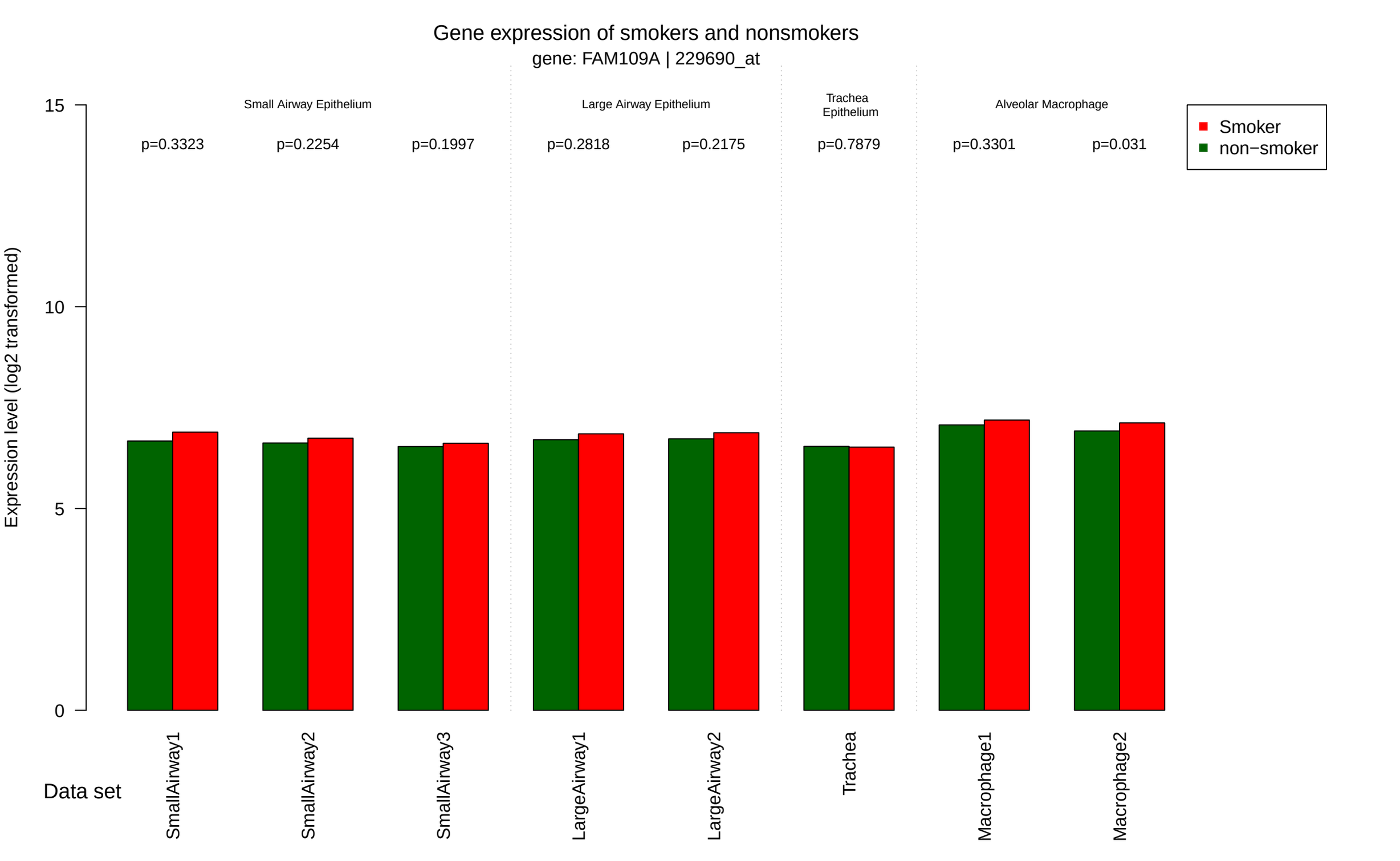
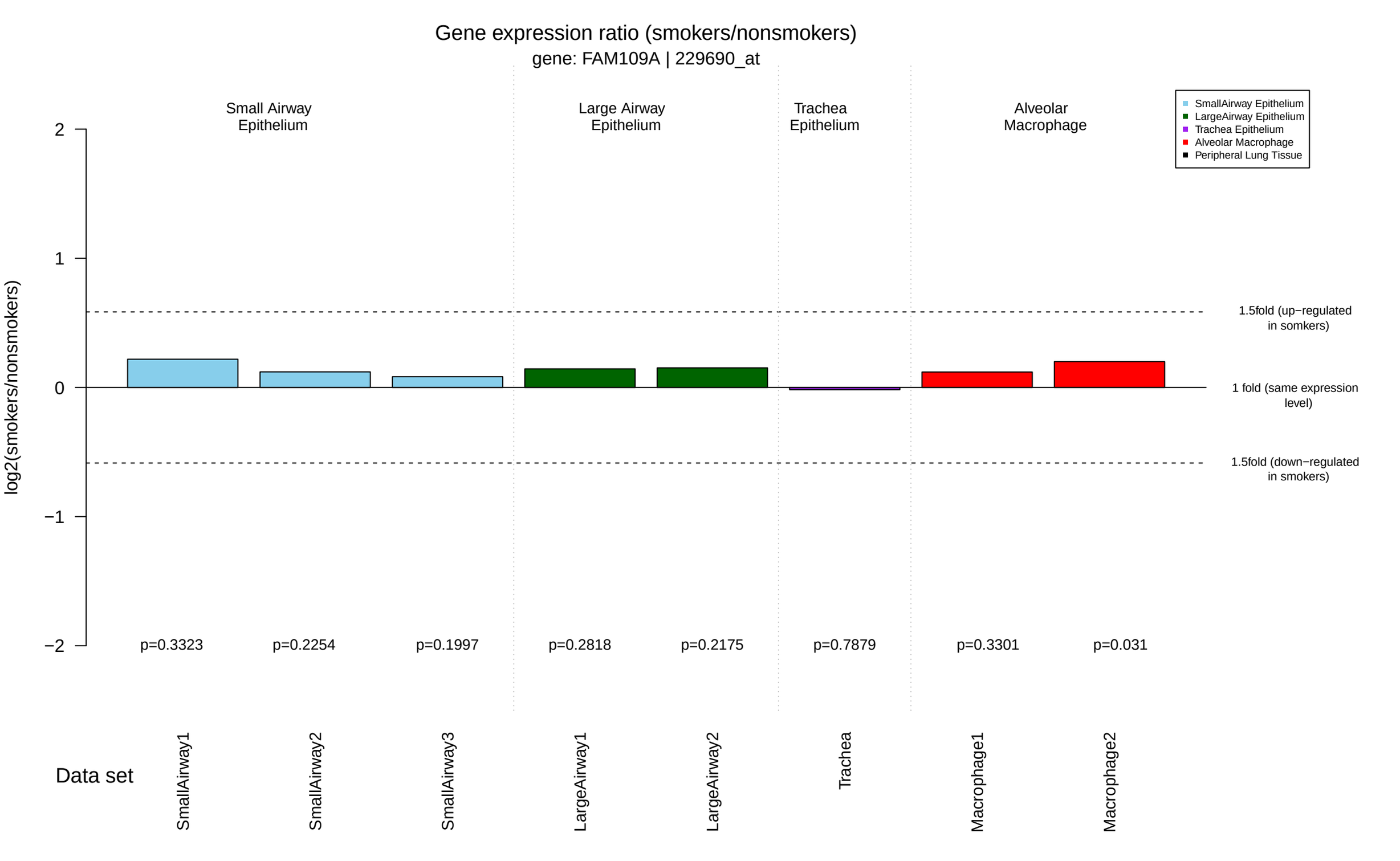
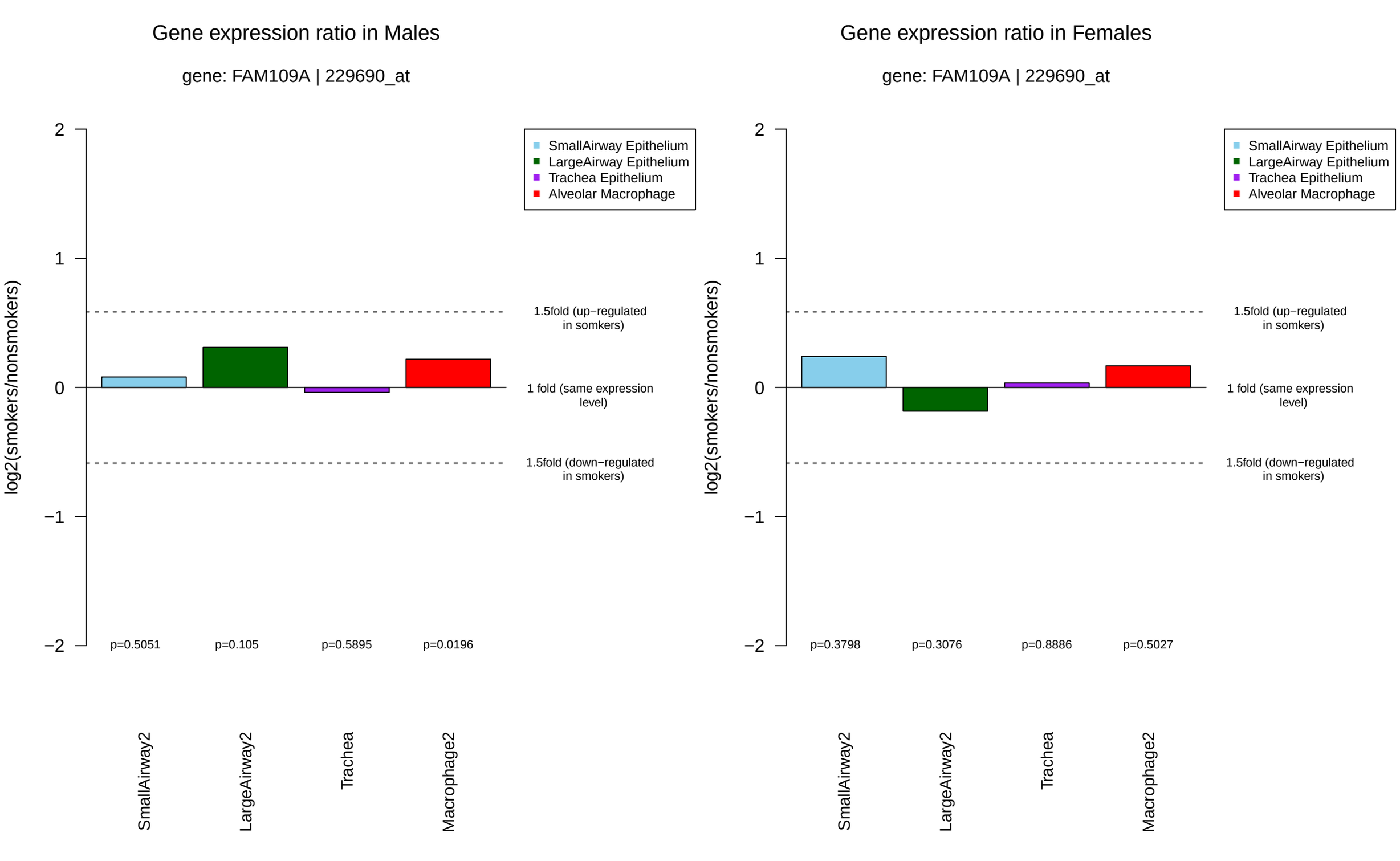
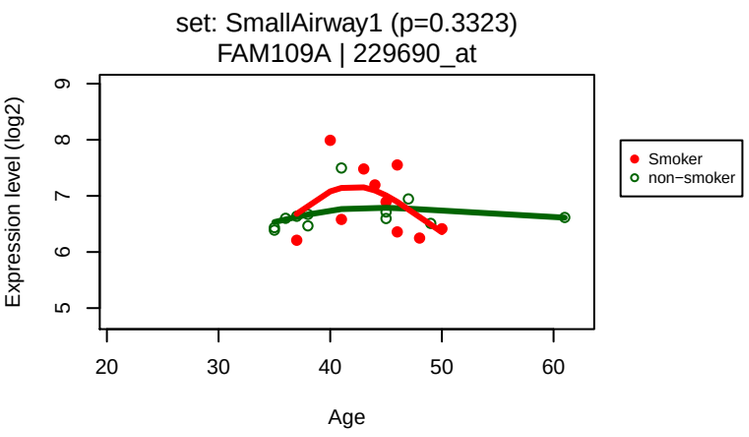
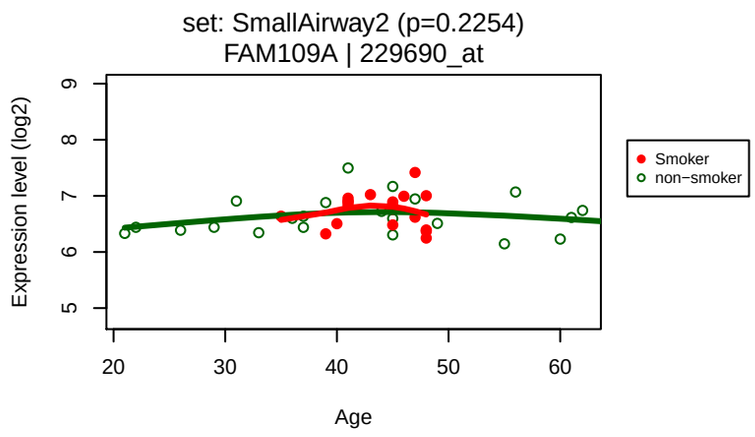
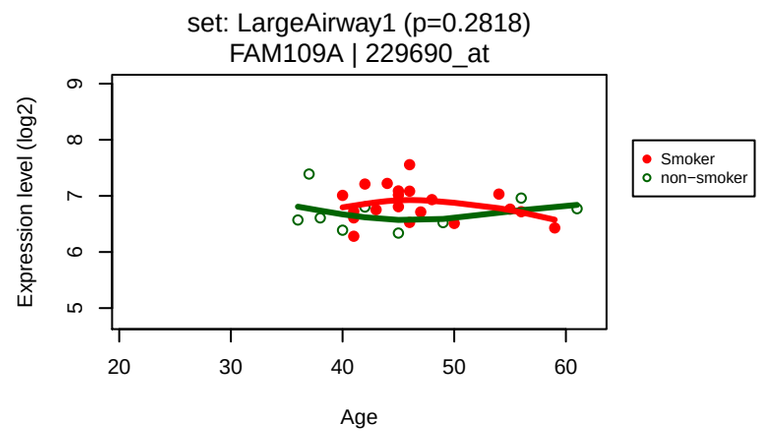

Gene expression of smokers and nonsmokers
gene: FAM109A | 229690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3323
p=0.2254
p=0.1997
p=0.2818
p=0.2175
p=0.7879
p=0.3301
p=0.031
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

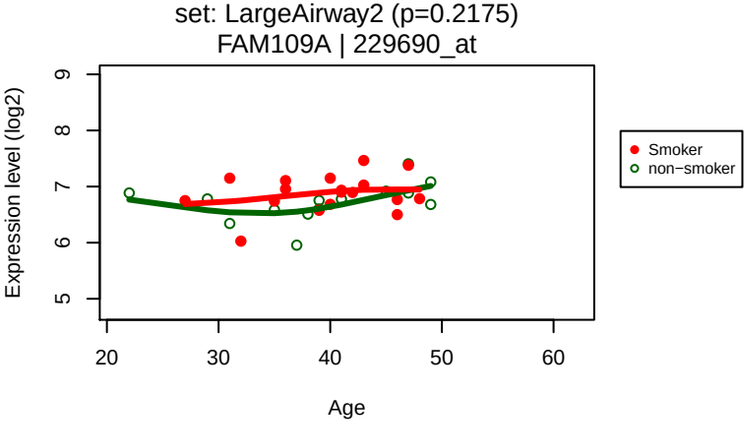

Gene expression ratio (smokers/nonsmokers)
gene: FAM109A | 229690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3323
p=0.2254
p=0.1997
p=0.2818
p=0.2175
p=0.7879
p=0.3301
p=0.031
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

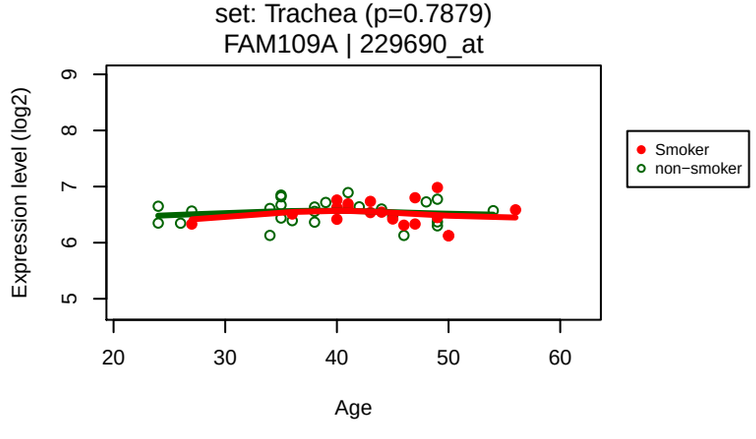

Gene expression ratio in Males
gene: FAM109A | 229690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5051
p=0.105
p=0.5895
p=0.0196
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FAM109A | 229690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3798
p=0.3076
p=0.8886
p=0.5027
SmallAirway2
LargeAirway2
Trachea
Macrophage2

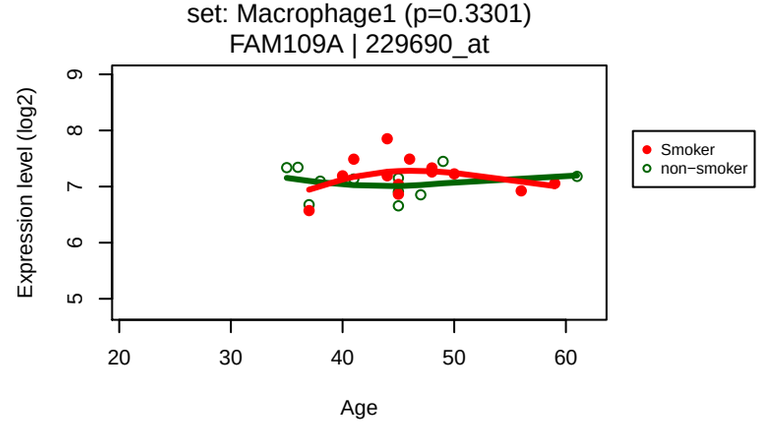

set: SmallAirway1 (p=0.3323)
FAM109A | 229690_at
Expression level (log2)
Age
Smoker
non−smoker

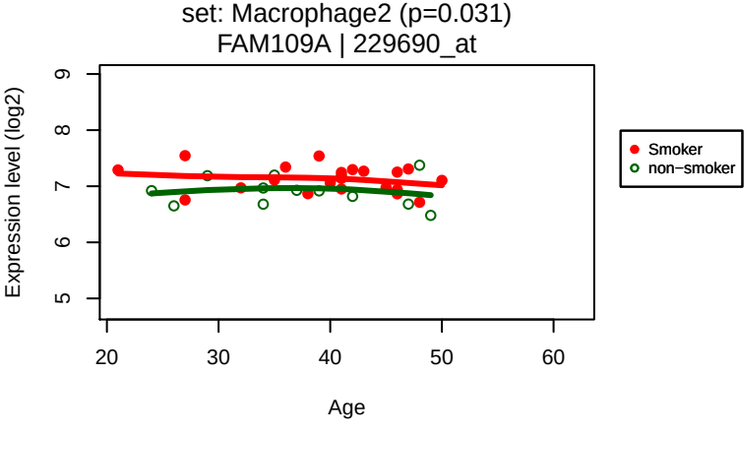

set: SmallAirway2 (p=0.2254)
FAM109A | 229690_at
Expression level (log2)
Age
Smoker
non−smoker

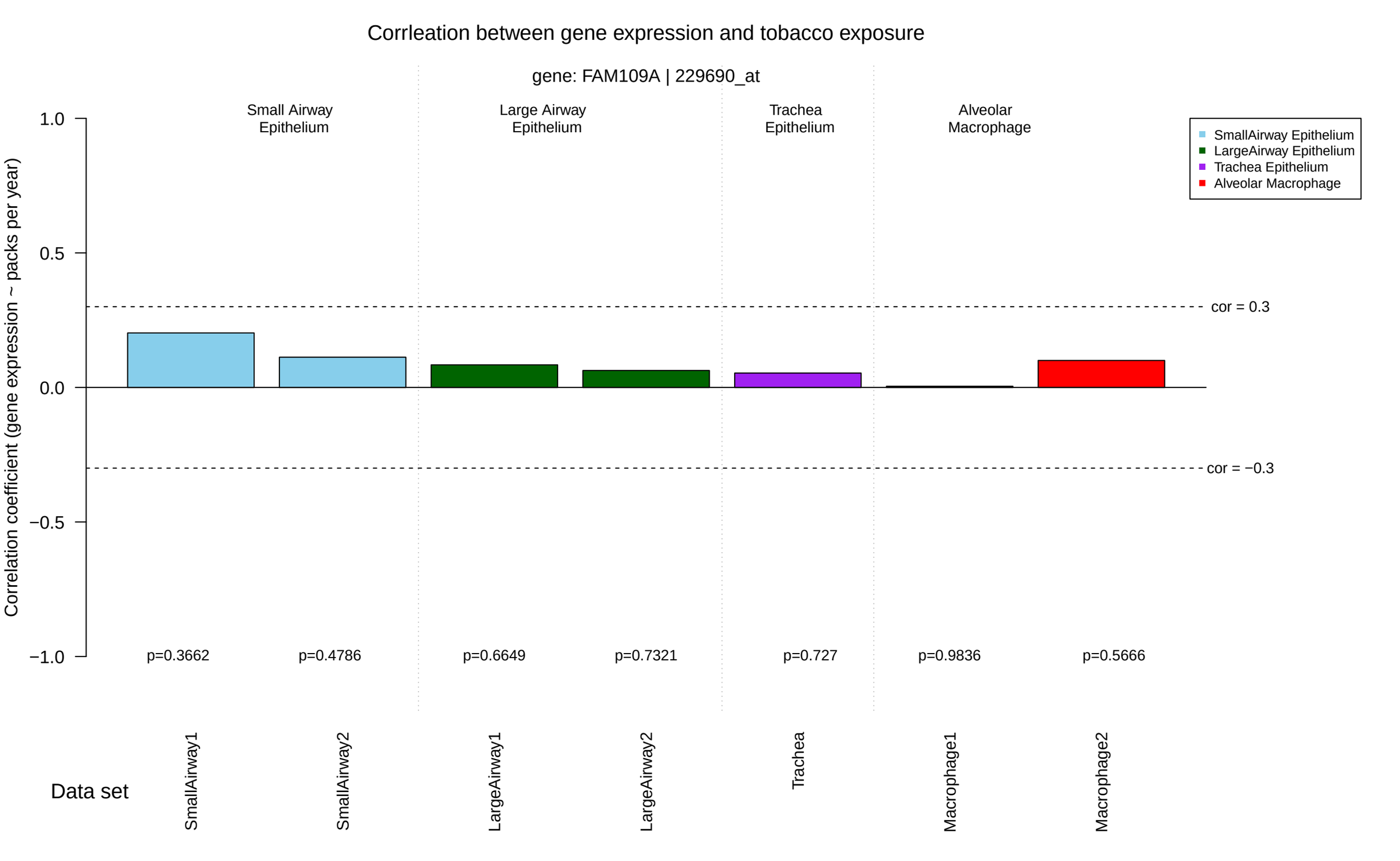

set: LargeAirway1 (p=0.2818)
FAM109A | 229690_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2175)
FAM109A | 229690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7879)
FAM109A | 229690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3301)
FAM109A | 229690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.031)
FAM109A | 229690_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FAM109A | 229690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3662
p=0.4786
p=0.6649
p=0.7321
p=0.727
p=0.9836
p=0.5666
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2